Condensed Matter Physics: Lecture 2

Preliminaries: Quantum & Statistical Foundations

Lecture Notes Series

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Contents

1 Quantum Mechanics

1.1 The Schrödinger Equation

The core non-relativistic differential equation governing single-particle quantum states in a spatial potential $V(\mathbf{r})$ is the time-independent Schrödinger Equation:

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r}) + V(\mathbf{r})\psi(\mathbf{r}) = E\psi(\mathbf{r}) \tag{1}$$

where $\hbar = h/2\pi$ is the reduced Planck constant, m is the particle mass, E is the energy eigenvalue, and $\psi(\mathbf{r})$ represents the spatial wave function.

1.2 Potential Well: $V(x) = -V_0\mathbf{sech}^2(x/l)$

To illustrate the distinction between discrete bound states and continuous scattering states in condensed matter systems, consider the smooth 1D potential well:

$$V(x) = -V_0\,\mathrm{sech}^2\left(\frac{x}{l}\right), \quad (V_0 > 0) \tag{2}$$

where V_0 defines the depth of the potential well and l parameterizes its spatial extension.

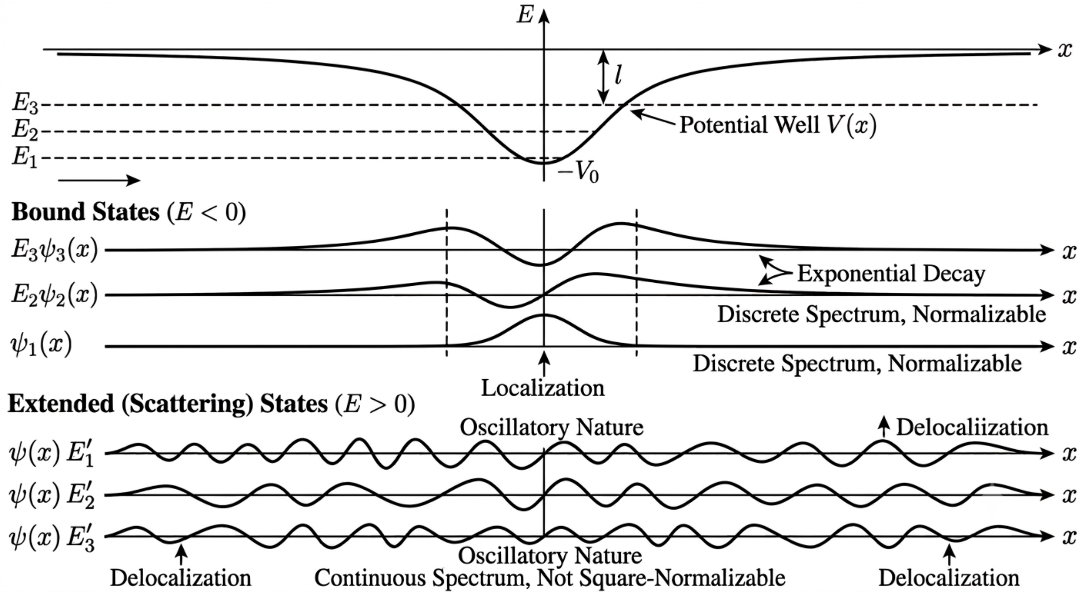

Figure 1: Schematic representation of the energy spectrum and wavefunctions for the $V(x) = -V_0\,\mathrm{sech}^2(x/l)$ potential well. Discrete bound states ($E < 0$) exhibit localized wavefunctions that decay exponentially at large distances. Extended scattering states ($E > 0$) form a continuous energy spectrum characterized by delocalized spatial oscillations.

1.2.1 Bound States ($E < 0$)

- **Energy Regime:** $E < 0$. The total energy is insufficient for the quantum particle to overcome the potential well depth as $x \to \pm\infty$.

- **Localization:** The wavefunction $\psi(x)$ is strictly **localized** inside and near the well region, exhibiting exponential decay in the classically forbidden regions:

$$\psi(x) \propto e^{-\kappa|x|} \quad \text{as } |x| \to \infty, \quad \text{where } \kappa = \frac{\sqrt{2m|E|}}{\hbar} \tag{3}$$

- **Spectrum:** Discrete energy levels E_n. The wavefunctions are square-integrable and strictly normalizable in coordinate space:

$$\int_{-\infty}^{\infty} |\psi_n(x)|^2 dx = 1 \tag{4}$$

- **Physical Context:** Represents localized electronic states around isolated atomic sites, dopants, or crystal defects.

1.2.2 Scattering / Extended States ($E > 0$)

- **Energy Regime:** $E > 0$. The particle moves freely across the spatial domain with asymptotic wavevector $k = \sqrt{2mE}/\hbar$.
- **Delocalization:** The wavefunctions exhibit sustained spatial oscillations throughout all space and do not decay to zero at infinity.
- **Spectrum:** Continuous spectrum. These states cannot be normalized via standard L^2-integrability over infinite space ($\int_{-\infty}^{\infty} |\psi_k(x)|^2 dx \to \infty$), requiring box normalization or periodic boundaries.

1.3 Scattering States & Born-von Karman (Periodic) Boundary Conditions

To handle continuous scattering states in bulk solid-state theory, we apply **Born-von Karman Periodic Boundary Conditions (PBC)** over a system of macro length L.

For a 1D crystal lattice or domain of length L, periodic boundary conditions demand:

$$\psi(x + L) = \psi(x) \tag{5}$$

Applying PBC to an extended plane wave $\psi(x) = Ae^{ikx}$:

$$e^{ik(x+L)} = e^{ikx} \implies e^{ikL} = 1 \tag{6}$$

This restricts the allowed continuous values of wavevector k into a discrete set of allowed grid points:

$$k_n = \frac{2\pi n}{L}, \quad n \in \mathbb{Z} \tag{7}$$

1.3.1 k-Space Properties in Condensed Matter

1. **Normalization:** Over the system volume $V = L^3$, plane waves are normalized as:

$$\psi_{\mathbf{k}}(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}} \tag{8}$$

2. k**-Space Grid Density:** The spacing between allowed momentum states is $\Delta k = \frac{2\pi}{L}$.

3. **Continuum Limit:** In the thermodynamic limit ($L \to \infty$), summations over discrete wavevectors transform smoothly into Brillouin zone integrals:

$$\sum_{\mathbf{k}} \longrightarrow \frac{V}{(2\pi)^3} \int d^3k \tag{9}$$

2 Statistical Mechanics

Statistical mechanics connects single-particle microscopic spectra $E(\mathbf{k})$ to macroscopic thermodynamic observables.

2.1 Classical Partition Function

In a classical system defined over a $6N$-dimensional phase space with positions $\mathbf{q} = (\mathbf{q}_1, \ldots, \mathbf{q}_N)$ and conjugate momenta $\mathbf{p} = (\mathbf{p}_1, \ldots, \mathbf{p}_N)$:

2.1.1 Canonical Ensemble Partition Function ($Z_{\mathbf{cl}}$)

$$Z_{\text{cl}} = \frac{1}{N!\, h^{3N}} \int d^{3N}q \, d^{3N}p \; e^{-\beta H(\mathbf{q},\mathbf{p})} \tag{10}$$

where:

- $\beta = \frac{1}{k_\text{B} T}$ represents the inverse thermodynamic temperature.
- $H(\mathbf{q}, \mathbf{p})$ is the classical system Hamiltonian.
- h^{3N} is the fundamental phase-space cell volume ensuring dimensionless state counting.
- $\frac{1}{N!}$ is the Gibbs correction factor for N indistinguishable particles.

2.1.2 Phase Space Probability Density

$$P(\mathbf{q}, \mathbf{p}) = \frac{e^{-\beta H(\mathbf{q},\mathbf{p})}}{Z_\text{cl}} \tag{11}$$

2.2 Quantum Partition Function

In quantum statistical mechanics, phase space integrations are replaced by operator traces over the Hilbert space.

2.2.1 Canonical Ensemble Partition Function ($Z_{\mathbf{qm}}$)

$$Z_\text{qm} = \text{Tr}\left(e^{-\beta \hat{H}}\right) = \sum_n \langle n | e^{-\beta \hat{H}} | n \rangle \tag{12}$$

If $\{|n\rangle\}$ is chosen as the exact **energy eigenbasis** of the Hamiltonian ($\hat{H}|n\rangle = E_n|n\rangle$):

$$Z_\text{qm} = \sum_n e^{-\beta E_n} \tag{13}$$

2.2.2 Density Matrix Operator ($\hat{\rho}$)

The thermal equilibrium state is completely described by the canonical density matrix operator:

$$\hat{\rho} = \frac{e^{-\beta \hat{H}}}{Z_\text{qm}} = \frac{1}{Z_\text{qm}} \sum_n e^{-\beta E_n} |n\rangle\langle n| \tag{14}$$

2.2.3 Ensemble Average of an Observable Operator ($\hat{A}$)

$$\langle \hat{A} \rangle = \mathrm{Tr}(\hat{\rho}\hat{A}) = \frac{\mathrm{Tr}\left(\hat{A}e^{-\beta\hat{H}}\right)}{Z_{\mathrm{qm}}} = \frac{1}{Z_{\mathrm{qm}}}\sum_n \langle n|\hat{A}|n\rangle e^{-\beta E_n} \tag{15}$$

2.3 Many-Particle Schrödinger Equation & Quantum Statistics

In condensed matter physics, systems typically consist of $N \sim 10^{23}$ interacting particles. Let $\mathbf{x}_i = (\mathbf{r}_i, \sigma_i)$ represent the combined spatial coordinate $\mathbf{r}_i$ and internal spin state σ_i of the i-th particle.

The many-body Hamiltonian for N particles interacting via pair potential $V_{\mathrm{int}}(\mathbf{r}_i - \mathbf{r}_j)$ in an external potential $V_{\mathrm{ext}}(\mathbf{r}_i)$ is:

$$\hat{H} = \sum_{i=1}^{N}\left[-\frac{\hbar^2}{2m}\nabla_i^2 + V_{\mathrm{ext}}(\mathbf{r}_i)\right] + \frac{1}{2}\sum_{i\neq j}^{N} V_{\mathrm{int}}(\mathbf{r}_i - \mathbf{r}_j) \tag{16}$$

The corresponding stationary many-particle Schrödinger equation reads:

$$\hat{H}\Psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) = E\Psi(\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N) \tag{17}$$

2.3.1 Identical Particles & Permutation Symmetry

In quantum mechanics, fundamental particles of the same species are fundamentally **indistinguishable**. Defining the transposition operator $\hat{P}_{ij}$ that exchanges particles i and j:

$$\hat{P}_{ij}\Psi(\ldots, \mathbf{x}_i, \ldots, \mathbf{x}_j, \ldots) = \Psi(\ldots, \mathbf{x}_j, \ldots, \mathbf{x}_i, \ldots) \tag{18}$$

Since the particles are indistinguishable, $[\hat{H}, \hat{P}_{ij}] = 0$. Furthermore, because $\hat{P}_{ij}^2 = \hat{I}$, the eigenvalues of $\hat{P}_{ij}$ must be $\lambda = \pm 1$. This leads to two fundamental quantum statistics symmetries:

2.3.2 1. Bosons (Symmetric Wavefunctions)

Particles with **integer spin** ($S = 0, 1, 2, \ldots$, e.g., phonons, photons, ^{4}He atoms) obey **Bose-Einstein statistics**. Their many-body wavefunction is completely symmetric under particle exchange:

$$\hat{P}_{ij}\Psi_{\mathrm{B}}(\ldots, \mathbf{x}_i, \ldots, \mathbf{x}_j, \ldots) = +\Psi_{\mathrm{B}}(\ldots, \mathbf{x}_i, \ldots, \mathbf{x}_j, \ldots) \tag{19}$$

* **Consequence:** Any number of bosons can occupy the same single-particle quantum state, driving physical phenomena such as **Bose-Einstein Condensation (BEC)** and **Superfluidity**.

2.3.3 2. Fermions (Antisymmetric Wavefunctions)

Particles with **half-integer spin** ($S = 1/2, 3/2, \ldots$, e.g., electrons, protons, neutrons) obey **Fermi-Dirac statistics**. Their many-body wavefunction is completely antisymmetric under particle exchange:

$$\hat{P}_{ij}\Psi_{\mathrm{F}}(\ldots,\mathbf{x}_i,\ldots,\mathbf{x}_j,\ldots) = -\Psi_{\mathrm{F}}(\ldots,\mathbf{x}_i,\ldots,\mathbf{x}_j,\ldots) \tag{20}$$

* **Pauli Exclusion Principle:** If two non-interacting fermions attempt to occupy the exact same single-particle state ϕ_α, setting $\mathbf{x}_i = \mathbf{x}_j = \mathbf{x}$ yields:

$$\Psi_{\mathrm{F}}(\ldots,\mathbf{x},\ldots,\mathbf{x},\ldots) = -\Psi_{\mathrm{F}}(\ldots,\mathbf{x},\ldots,\mathbf{x},\ldots) \implies \Psi_{\mathrm{F}} = 0 \tag{21}$$

No two identical fermions can occupy the same quantum state.

2.3.4 When Do Quantum Statistics Become Important?

Classical Maxwell-Boltzmann statistics fail, and quantum statistics become strictly mandatory, when the spatial extent of individual particle wavepackets overlaps significantly with neighboring particles.

Thermal de Broglie Wavelength ($\lambda_{\mathbf{th}}$) At thermodynamic temperature T, a particle of mass m has an average thermal momentum $p_{\mathrm{th}} \sim \sqrt{mk_{\mathrm{B}}T}$. The associated quantum mechanical spatial wavepacket spread is defined by the **thermal de Broglie wavelength**:

$$\lambda_{\mathrm{th}} = \sqrt{\frac{2\pi\hbar^2}{mk_{\mathrm{B}}T}} = \frac{h}{\sqrt{2\pi mk_{\mathrm{B}}T}} \tag{22}$$

Quantum Degeneracy Condition For a system with particle number density $n = N/V$, the mean interparticle distance is $d \sim n^{-1/3}$. The crossover into the quantum regime is governed by the dimensionless **degeneracy parameter** $n\lambda_{\mathrm{th}}^3$:

1. **Non-Degenerate / Classical Regime ($n\lambda_{\mathbf{th}}^3 \ll 1$):**
   The mean spacing between particles is vastly larger than their thermal wavelength ($d \gg \lambda_{\mathrm{th}}$). Wavepackets do not overlap, indistinguishability is negligible, and the system behaves as classical billiard balls obeying Maxwell-Boltzmann statistics.

2. **Degenerate / Quantum Regime ($n\lambda_{\mathbf{th}}^3 \gtrsim 1$):**
   The thermal wavelength becomes comparable to or larger than the interparticle distance ($\lambda_{\mathrm{th}} \gtrsim d$). The spatial wavefunctions overlap heavily, making particle indistinguishability and exchange symmetry dominant (Fermi-Dirac or Bose-Einstein statistics).

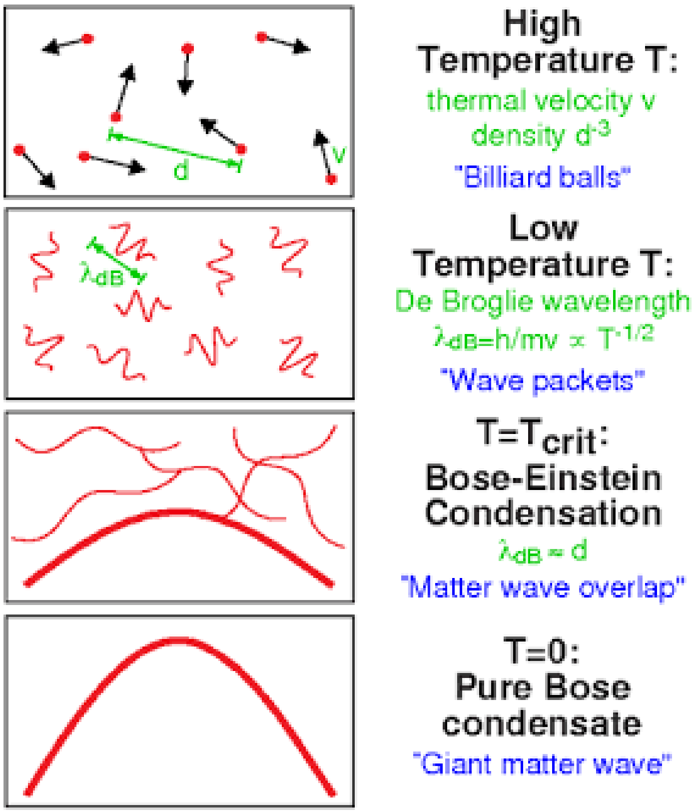

Figure 2: The transition from classical behavior to quantum degeneracy and Bose-Einstein condensation. Left: At high temperatures ($n\lambda_{\rm th}^3 \ll 1$), wavepackets are small ($d \gg \lambda_{\rm th}$) and particles act classically. Middle: Near the critical temperature $T \approx T_c$, thermal wavepackets expand and overlap ($d \approx \lambda_{\rm th}$). Right: At $T = 0$, individual wavepackets merge into a single macroscopic quantum wavefunction $\Psi(\mathbf{r})$.